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Paper 1: Model-Informed Drug Development of Subcutaneous Nivolumab: Comparison of Pharmac

Bibliographic Information

- **Title:** Model-Informed Drug Development of Subcutaneous Nivolumab: Comparison of Pharmacokinetic Analysis Methodologies Using Clinical Trial Simulation
- **Authors:** Zhao Y, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2025;14(12):e70120
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Quick Take

Pharmacometricians designing PK non-inferiority studies for subcutaneous biologics should read this paper. Using clinical trial simulation, the authors validate that a pre-specified popPK model with NONMEM's \$PRIOR subroutine yields more accurate exposure estimates than conventional NCA and matches pooled analysis, providing the methodological foundation for the CheckMate 67T s.c. nivolumab registration.

Executive Summary

This paper provides the methodological validation for the model-based determination of the co-primary PK exposure endpoints (Cavgd28 and Cminss) in CheckMate 67T, the pivotal phase III study comparing subcutaneous nivolumab 1200 mg q4w to intravenous nivolumab 3 mg/kg q2w. Using stochastic simulation-estimation across 87 replicate clinical trials that mirror the CheckMate 67T design in resampled RCC patients, the authors compared three analysis approaches: NONMEM's \$PRIOR subroutine, pooled popPK analysis with historical data, and NCA. Both model-based approaches produced consistent exposure estimates (differences <2% for Cavgd28, Cmax1, and Cmind28; <5% for Cminss) and were more accurate than NCA with conventional sampling, which overestimated i.v. AUC due to biphasic elimination and underestimated s.c. AUC due to absorption-flattened profiles. The work establishes \$PRIOR as a feasible, pre-specified, and robust alternative to pooled analysis for leveraging extensive prior PK knowledge in pivotal PK non-inferiority assessments, with important implications for reducing PK sampling burden in clinical trials.

Scientific Context & Motivation

The development of subcutaneous biologics requires demonstration of PK non-inferiority to the approved i.v. regimen. Traditional approaches rely on NCA-derived exposure metrics, which require intensive PK sampling and can be biased by non-random dropout and by the relationship between sampling schedule and the shape of the concentration-time profile. This paper addresses

a key methodological gap: whether a model-based approach using prior information can robustly support a pivotal PK non-inferiority endpoint. It challenges the paradigm that primary PK endpoints in registration trials must be derived from observed data alone, and demonstrates that leveraging extensive historical i.v. PK data (18 studies, N=3488) plus phase I/II s.c. absorption data through the \$PRIOR subroutine provides accurate and precise exposure estimates. The work also addresses practical considerations for \$PRIOR implementation, including specification of prior distributions (multivariate normal for fixed effects, inverse-Wishart for IIV) and the importance of a well-validated reference model.

✂ Methodological Snapshot

The study used a stochastic simulation-estimation (SSE) approach. A pre-specified two-compartment popPK model with time-varying clearance and first-order s.c. absorption was developed from 18 historical i.v. studies (N=3488) and CheckMate 8KX s.c. data (N=66). One hundred datasets of 400 resampled RCC patients (200 per arm) were simulated under the CheckMate 67T design, with population parameters sampled from the joint uncertainty distribution. Each simulated dataset was re-analyzed using (1) \$PRIOR in NONMEM with priors on structural parameters (CL, VC, VP, Q, EMAX, HILL, T50, KA, F) and IIV (CL, VC, VP, EMAX), and (2) pooled analysis with historical data. EBE-derived exposures (Cavgd28, Cmax1, Cmind28, Cavgss, Cminss, Cmaxss) were compared to NCA under three sampling schemes. Bias and precision were assessed against true simulated values.

☒ Structural Model Breakdown

Two-compartment disposition model with time-varying clearance common to i.v. and s.c. routes: central compartment (VC) and peripheral compartment (VP) connected by inter-compartmental clearance (Q). Clearance decreases over time from baseline CL toward $CL \times (1 - EMAX)$, following a Hill function with T50 (time to 50% of maximum change) and HILL coefficient. For s.c. administration, an additional absorption compartment with first-order rate constant KA and bioavailability F feeds the central compartment. IIV was estimated on CL, VC, VP, and EMAX. The model was pre-specified with covariates relevant to the RCC population (e.g., body weight, eGFR, albumin, ECOG PS, tumor type effects as previously established).

Detailed Methodological Analysis

Modeling Approach

Two-compartment disposition model with time-varying clearance (CL, VC, VP, Q, EMAX, T50, HILL) and first-order s.c. absorption (KA, F). Time-varying clearance was modeled as $CL(t) = CL \times (1 - EMAX \times t^{HILL} / (T50^{HILL} + t^{HILL}))$. Analysis approaches: (1) \$PRIOR subroutine in NONMEM with priors on structural parameters and IIV; (2) pooled popPK analysis combining simulated data with historical data; (3) NCA with three sampling schemes (conventional, intensive q12h, intensive q24h). Software: NONMEM 7.4 (FOCE-I), Perl-speaks-NONMEM 4.9.0, R 4.0.2.

Data Sources

Historical i.v. data from 18 nivolumab monotherapy studies (N=3488 patients) across NSCLC, melanoma, RCC, SCCHN, urothelial, and gastric cancers; s.c. data from CheckMate 8KX (N=66 patients; those without rHuPH20 excluded); simulated CheckMate 67T data with 400 patients per trial (200 s.c. 1200 mg q4w, 200 i.v. 3 mg/kg q2w), 100 replicate trials, 87 with valid OMEGA. PK sampling for popPK mimicked CheckMate 67T: cycle 1 days 1, 4, 8, 15, 22; pre-dose at cycles 2-5. NCA sampling: i.v. cycle 1 at 0.5, 4, 8, 24, 72, 168, 336, 336.5, 504, 672 h; s.c. at 24, 72, 168, 336, 480, 672 h; cycle 5 at 3360 h.

Estimation Methods

First-order conditional estimation with interaction (FOCE-I) in NON-MEM 7.4. *PRIOR* implementation: *multivariate normal prior for fixed effects* (THETAP, THETAPV), *multivariate inverse – Wishart prior for IIV* (OMEGAP). Population parameters for simulation were sampled from a multivariate normal distribution using the variance-covariance matrix of parameter uncertainty.

Model Evaluation

Prediction-corrected VPC (pcVPC) for the \$PRIOR analysis; comparison of re-estimated parameters to true simulation values; assessment of bias and precision of exposure metrics (Cavgd28, Cmax1, Cmind28, Cavgss, Cminss, Cmaxss) via geometric mean distributions across simulations; comparison of s.c./i.v. GMRs across approaches.

Covariate Analysis

The reference model included only covariates relevant to the second-line RCC population. Covariate effects were not included in the \$PRIOR specification (no priors on covariate effects), allowing them to be estimated from the study data. The largest discrepancy between \$PRIOR and pooled estimates was the eGFR effect on CL, attributed to eGFR imbalance in resampling, with negligible impact (<15%).

Statistical Rigor Assessment

The SSE framework is statistically rigorous: 100 replicate trials with resampled patients and parameter uncertainty sampling provide a comprehensive assessment of bias and precision. The use of 87 valid OMEGA matrices (87% success rate) is reasonable but suggests some instability in IIV estimation. The comparison of geometric mean distributions and GMRs across approaches directly addresses the PK non-inferiority question. However, the analysis does not formally test for differences between \$PRIOR and pooled approaches (e.g., via confidence intervals on the differences), and the simulation-estimation circularity limits inference about model misspecification. The eGFR imbalance issue highlights the importance of covariate balance in resampling, though the authors appropriately assessed its impact.

☒ Key Findings

Across 87 simulated clinical trials, both \$PRIOR and pooled popPK approaches yielded nearly identical exposure estimates: differences in geometric means were <2% for Cavgd28, Cmax1, and Cmind28, and <5% for Cminss. Model-based analyses provided more accurate AUC estimates than NCA with conventional sampling: NCA overestimated i.v. AUC (due to the rapid distribution phase being poorly captured by sparse sampling) and underestimated s.c. AUC (due to the absorption-flattened elimination phase). NCA with intensive sampling (q12h or q24h) approached model-based accuracy but is impractical in clinical trials. The GMRs (s.c./i.v.) for all exposure metrics were consistent between \$PRIOR and pooled approaches. Parameter estimates from both approaches were similar to the ‘true’ simulation values; the largest discrepancy was the eGFR effect on CL, attributed to sampling imbalance, but its impact was negligible (<15% effect size). The pcVPC confirmed that the \$PRIOR-based analysis adequately described the simulated data.

🔗 Clinical & Regulatory Implications

The validated model-based approach was used to determine Cavgd28 and Cminss as co-primary endpoints for PK non-inferiority in CheckMate 67T, supporting the regulatory approval of s.c. nivolumab 1200 mg q4w. The approach reduces PK sampling burden (5 samples in cycle 1 plus pre-dose troughs vs intensive NCA sampling), improving patient convenience and site feasibility. The demonstration that s.c. Cmaxss remains below the 10 mg/kg i.v. q2w safety margin supports the safety of the s.c. regimen. This MIDD framework may serve as a template for future s.c. biologics development and regulatory submissions, potentially influencing FDA and EMA expectations for PK non-inferiority assessment.

Strengths & Limitations

Strengths

- Rigorous stochastic simulation-estimation framework with 100 replicate trials (87 with valid positive-definite OMEGA matrices), pairing resampled RCC patients with parameter sets sampled from the joint uncertainty distribution
- Head-to-head comparison of three analysis methodologies (\$PRIOR, pooled popPK, NCA) under realistic trial conditions mirroring the CheckMate 67T design
- Use of a pre-specified, well-validated reference model built on extensive historical data (18 i.v. studies, N=3488; s.c. data from CheckMate 8KX)
- Clinically meaningful endpoint assessment including distributions of s.c./i.v. GMRs for PK non-inferiority
- Practical demonstration that \$PRIOR can replace pooled analysis, avoiding re-analysis of historical data and reducing run times

Limitations (Acknowledged by Authors)

- Only 87 of 100 simulated datasets had a valid (positive-definite) OMEGA matrix and were used, potentially introducing selection bias

- Imbalance in eGFR distribution during patient resampling led to the largest parameter discrepancy (eGFR effect on CL), though impact was deemed negligible
- NCA results are sensitive to the pre-specified sampling schedule and the shape of the concentration-time profile

Limitations (Expert Review)

- Simulation-estimation circularity: the ‘true’ model is the same model used for estimation, so the assessment does not test robustness to structural model misspecification
- The \$PRIOR approach assumes the historical prior is relevant to the target population (ccRCC); the prior included multiple tumor types, and the s.c. prior data were limited (n=66)
- No evaluation of the impact of non-random dropout or missing data on the model-based vs NCA comparison, despite this being cited as an advantage of the model-based approach
- The inverse-Wishart prior specification for OMEGA can be sensitive to prior degrees of freedom; sensitivity analyses were not reported
- The comparison of NCA sampling schemes was limited to the described schedules; the optimal NCA sampling design was not systematically explored

Generalizability

The findings are likely generalizable to other monoclonal antibodies with well-characterized i.v. PK and a validated popPK model, particularly where s.c. absorption is slow relative to elimination (flip-flop kinetics). However, the approach requires a substantial historical PK database and a robust reference model; for novel mechanisms or limited prior data, the \$PRIOR approach would need additional validation. The regulatory acceptance of this approach in CheckMate 67T may set a precedent for future s.c. biologics development.

Key Equations

Time-varying clearance model

$$CL(t) = CL_{\text{pop}} \times \left(1 - \frac{E_{\text{MAX}} \times t^{\text{HILL}}}{T_{50}^{\text{HILL}} + t^{\text{HILL}}} \right) \times e^{\eta_{CL}}$$

Describes the gradual decrease in nivolumab clearance over time, where EMAX is the maximum fractional reduction, T50 is the time to 50% of EMAX, and HILL is the Hill coefficient.

Central compartment mass balance

$$\frac{dA_c}{dt} = - \left(\frac{CL(t)}{V_c} + \frac{Q}{V_c} \right) \cdot A_c + \frac{Q}{V_p} \cdot A_p + \text{Input}$$

Mass balance for the central compartment in the two-compartment disposition model common to i.v. and s.c. routes.

Peripheral compartment mass balance

$$\frac{dA_p}{dt} = \frac{Q}{V_c} \cdot A_c - \frac{Q}{V_p} \cdot A_p$$

Mass balance for the peripheral compartment, with inter-compartmental clearance Q connecting central and peripheral volumes.

Subcutaneous absorption compartment

$$\frac{dA_a}{dt} = -K_A \cdot A_a, \text{Input} = K_A \cdot A_a \cdot F$$

First-order absorption from the subcutaneous depot with rate constant K_A and bioavailability F feeding the central compartment.

Time-averaged concentration endpoint

$$C_{avgd28} = \frac{AUC_{0-28d}}{28}$$

Definition of the co-primary exposure endpoint C_{avgd28} , the time-averaged serum concentration over the first 28 days.

Figures & Tables

- **Figure 1:** General workflow of the MIDD approach: historical i.v. and s.c. data to pre-specified popPK model, clinical trial simulation of CheckMate 67T, then comparison of \$PRIOR, pooled popPK, and NCA analyses.
 - *Significance:* Provides the conceptual framework for the entire study and illustrates how prior information and clinical trial simulation are integrated into the model-based PK non-inferiority assessment.
- **Figure 2:** Median (90% PI) of simulated geometric mean concentration-time profiles by route of administration (s.c. 1200 mg q4w vs i.v. 3 mg/kg q2w) across simulated clinical trials.
 - *Significance:* Demonstrates the wider distribution for s.c. vs i.v. due to additional absorption parameter variability, and shows that predicted s.c. C_{maxss} remains well below the 10 mg/kg i.v. q2w safety margin.
- **Figure 3:** Distribution of nivolumab exposures (C_{avgd28} , C_{max1} , C_{mind28}) by administration route and analysis method, without parameter uncertainty.
 - *Significance:* Shows that \$PRIOR and pooled popPK produce nearly identical exposure distributions, and that NCA-24 results are comparable to model-based estimates.
- **Figure 4:** Distribution of estimated geometric mean nivolumab exposures (C_{avgd28} , C_{max1} , C_{mind28}) by administration route and analysis method, with parameter uncertainty across 87 simulations.
 - *Significance:* Confirms that the similarity between \$PRIOR and pooled approaches holds when parameter uncertainty is propagated, supporting the robustness of the model-based approach.
- **Figure 5:** Comparison of simulated geometric mean ratios (s.c./i.v.) of nivolumab exposures (C_{avgd28} , C_{max1} , C_{mind28} , C_{minss}) by popPK analysis approach and NCA, with uncertainty.

- *Significance*: Directly addresses the PK non-inferiority question by showing consistent GMRs between \$PRIOR and pooled approaches, and highlighting the lower GMR for Cavgd28 from conventional NCA.
 - **Table 1**: Baseline patient demographics for the s.c. (CheckMate 8KX, n=66) and i.v. (18 historical studies, n=3488) populations used to develop the reference popPK model.
 - *Significance*: Documents the characteristics of the prior data, including age, body weight, albumin, eGFR, sex, race, ECOG PS, and tumor type, supporting the relevance of the prior to the RCC target population.
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Code & Reproducibility Assessment

NONMEM control streams for the pre-specified popPK model and the \$PRIOR implementation were provided as supplementary files (File S1, File S2). However, the simulation datasets, full analysis code, and NCA analysis scripts were not publicly deposited. The analysis used NONMEM 7.4, Perl-speaks-NONMEM 4.9.0, and R 4.0.2.

Supplementary Materials

Supplementary materials include Table S1 (list of 18 historical i.v. nivolumab studies), Table S2 (population PK parameter estimates for the reference s.c./i.v. model), Table S3 (comparison of parameter estimates from a representative simulated dataset using \$PRIOR vs pooled analysis vs true values), Figure S1 (prediction-corrected VPC for the \$PRIOR analysis), File S1 (NONMEM control stream for the pre-specified popPK model), and File S2 (NONMEM control stream with \$PRIOR specification).

Future Directions

Future work could evaluate the robustness of the \$PRIOR approach under structural model misspecification (e.g., different time-varying clearance functions, alternative absorption models). Sensitivity analyses of prior specifications (e.g., inverse-Wishart degrees of freedom, down-weighting of prior information) would strengthen confidence. The approach could be extended to other biologics and to endpoints beyond PK (e.g., model-based efficacy bridging). Additionally, comparison with full Bayesian hierarchical approaches (e.g., Stan, NONMEM Bayesian) would position \$PRIOR within the broader landscape of prior-informed analysis. Finally, the impact of informative dropout on model-based vs NCA exposure estimates warrants formal investigation.

Expert Commentary

This paper represents a significant milestone in the regulatory application of MIDD: using prior information via \$PRIOR to support primary PK endpoints in a pivotal phase III study. From a senior perspective, several elements are noteworthy. First, the pre-specification of the analysis model and prior distributions is exemplary and aligns with regulatory expectations for reducing bias. Second, the clinical trial simulation framework — pairing resampled target-population patients with parameter uncertainty samples — is a rigorous template for assessing analysis

robustness before trial completion. Third, the demonstration that conventional NCA is biased for both i.v. (overestimation due to distribution phase) and s.c. (underestimation due to flip-flop kinetics) exposures is an important educational point: NCA is not a gold standard, merely a model-free approximation whose accuracy depends on sampling density. The practical implication is significant: model-based approaches can reduce PK sampling burden in pivotal trials, improving patient experience and site feasibility. However, I would caution that the success of \$PRIOR hinges on the quality and relevance of the prior model — this is not a free lunch. The 87/100 valid OMEGA rate also reminds us that IIV matrix estimation remains fragile; practitioners should plan for such contingencies. Overall, this is a landmark example of how pharmacometrics can de-risk drug development and shape regulatory strategy.

Bottom Line

For practicing pharmacometricians, this paper validates a practical pathway for using prior information in pivotal PK non-inferiority assessments: a pre-specified popPK model with \$PRIOR can replace both NCA and pooled analysis, yielding more accurate exposure estimates with less intensive PK sampling. The key requirements are a well-validated reference model, appropriate specification of prior distributions (multivariate normal for fixed effects, inverse-Wishart for IIV), and demonstration of robustness via clinical trial simulation before trial results are available. This MIDD approach was successfully applied to support the CheckMate 67T registration of subcutaneous nivolumab.

☒ Figures

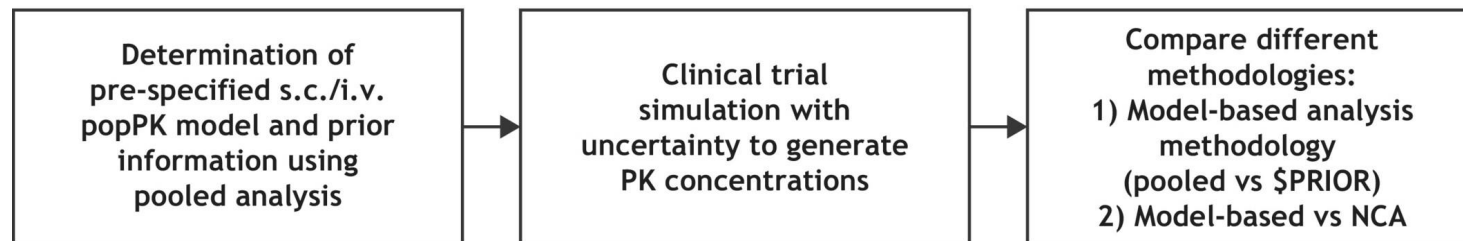


Figure 1: General workflow. i.v., intravenous; NCA, non-compartmental analysis; PK, pharmacokinetic; pop, population; s.c., subcutaneous.

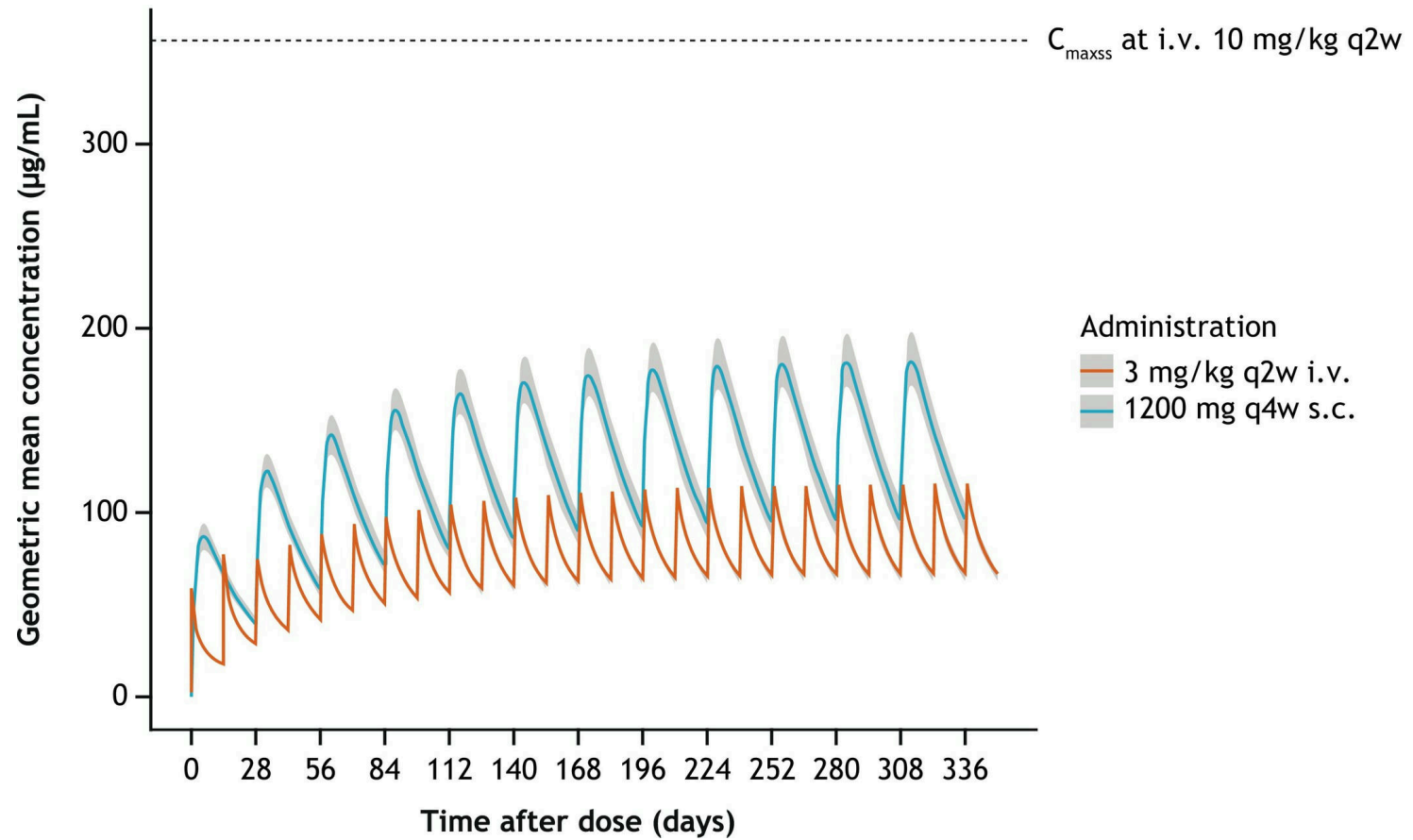


Figure 2: Median (90% PI) of simulated geometric mean concentration–time profile by route of administration. Solid line and the shaded area represent the median (90% PI) o

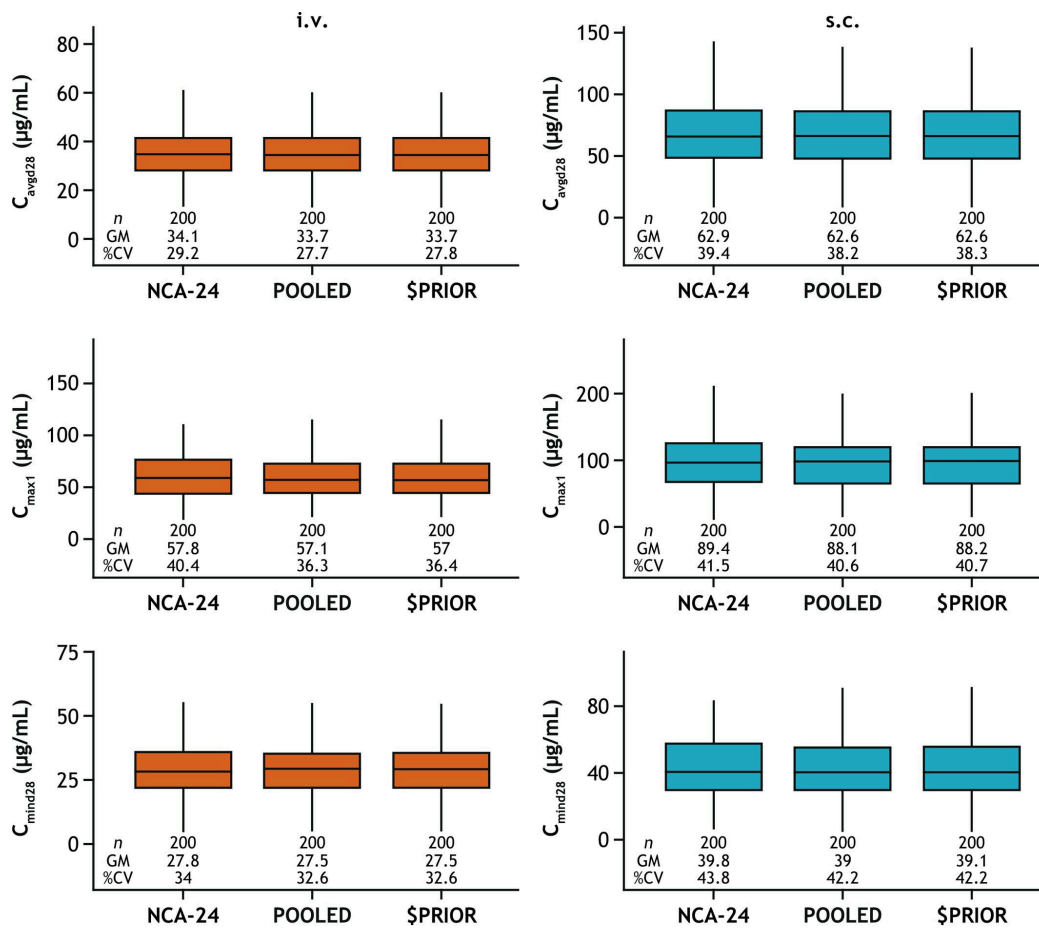


Figure 3: Distribution of nivolumab exposures (C_{avgd28} , C_{max1} , C_{mind28}) by administration route and analysis method (without uncertainty). PRIOR, PRIOR subroutine in NONMEM

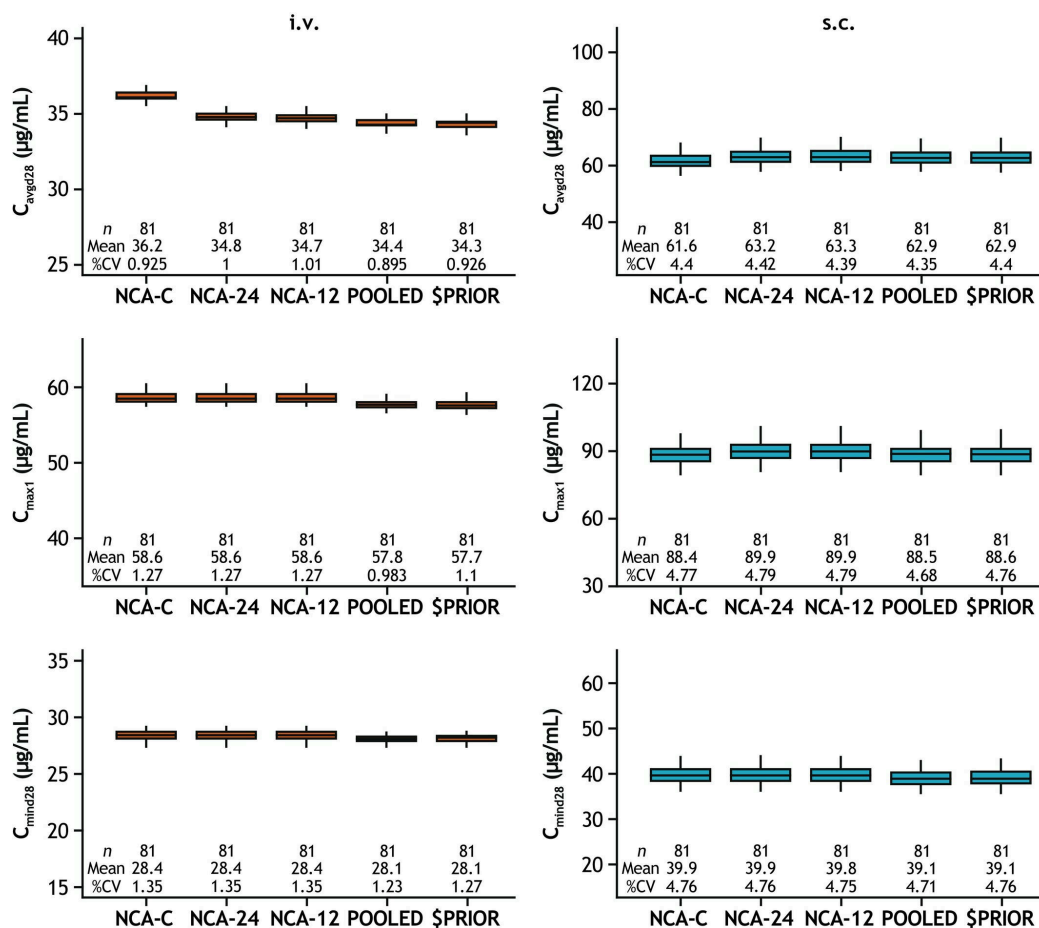


Figure 4: Distribution of estimated geometric mean nivolumab exposures (Cavgd28, Cmax1, Cmind28) by administration route and analysis method (with uncertainty).
PRIOR, PRIO

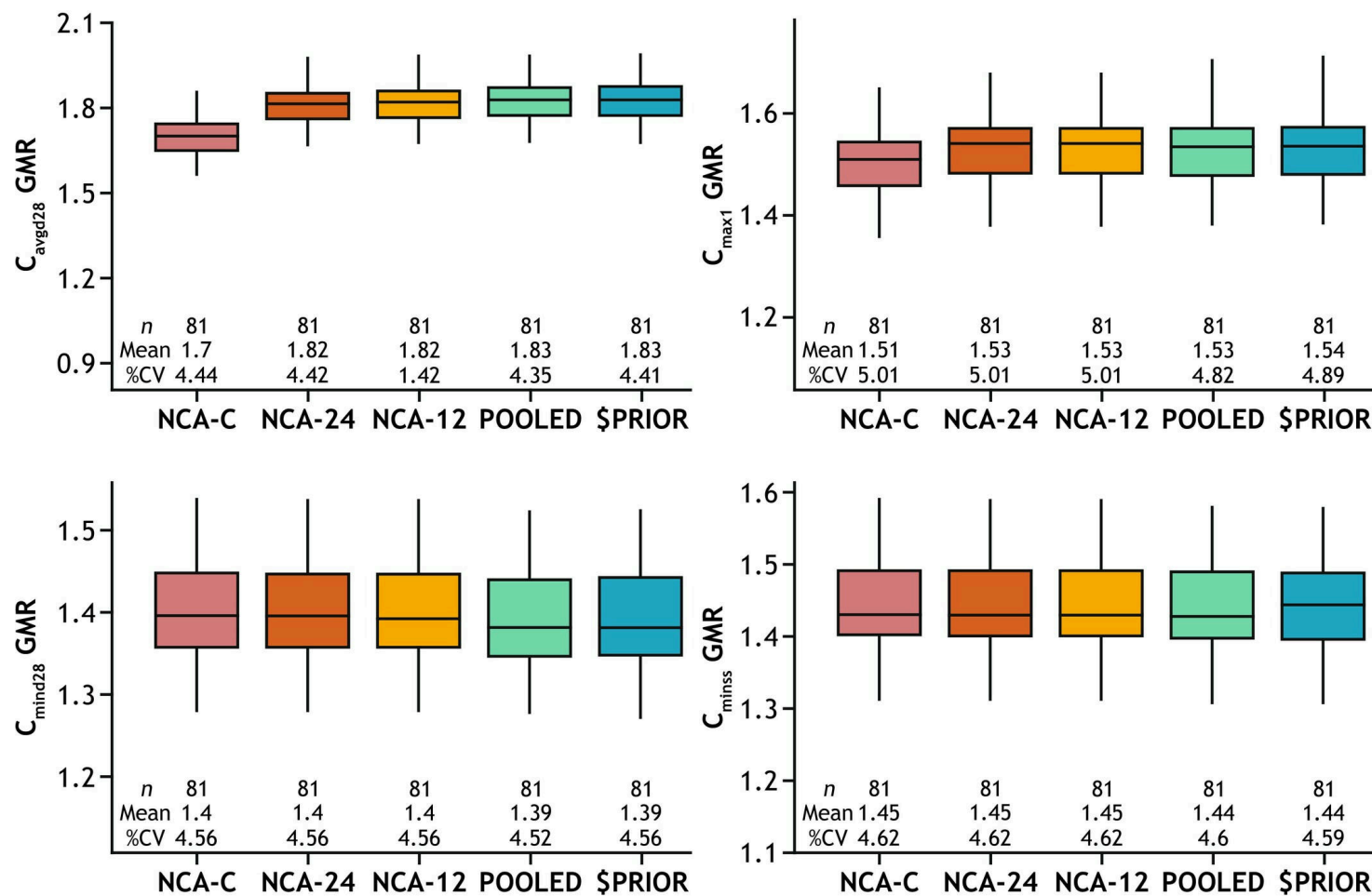


Figure 5: Comparison of simulated geometric mean ratios (s.c./i.v.) of nivolumab (Cavgd28, Cmax1, Cmind28, Cminss) by popPK analysis approach and NCA (with uncertainty). PRI